Wavelet Siamese Network With Semi-Supervised Domain Adaptation for Remote Sensing Image Change Detection

Fengchao Xiong, *Member, IEEE*, Tianhan Li, Yi Yang, Jun Zhou, *Senior Member, IEEE*, Jianfeng Lu, *Member, IEEE*, and Yuntao Qian, *Senior Member, IEEE*

***Abstract*— Change detection is a crucial technique in remote sensing image analysis and faces challenges, such as background complexity and appearance shift, resulting in incomplete change boundaries and pseudochanges. This article introduces a novel wavelet Siamese network with semi-supervised domain adaptation (DA) to address these issues, named WS-Net++. WS-Net++ establishes spatial–frequency interactions between bitemporal images to enhance the completeness of the change boundaries. The spatial-domain interaction highlights the pixelwise differences. The frequency-domain interaction first adaptively adjusts the contributions from different frequency components based on image context. Within-frequency and between-frequency interactions are further constructed to capture the frequency-domain differences, enabling the adaptive and effective handling of both overall and subtle changes. In addition, WS-Net++ employs a semi-supervised DA strategy to mitigate the appearance shifts between bitemporal images. By categorizing regions into changed, unchanged, and regions of no interest in a semi-supervised manner, the network minimizes intraclass discrepancies within unchanged regions and maximizes interclass discrepancies between changed regions, reducing the domain gap. Experimental results on the LEVIR-CD, WHU-CD, and CLCD datasets demonstrate that our WS-Net++ outperforms alternative methods, achieving the $F1$ scores of 91.31%, 94.52%, and 79.77%, respectively. The code and models will be publicly available at https://github.com/JiTaiTai/WS-Net_Plus for reproducible research.**

***Index Terms*— Change detection, domain adaptation (DA), frequency-domain analysis, remote sensing image.**

I. Introduction

CHANGE detection stands out as one of the highly successful applications in the field of remote sensing. By comparing variations in areas observed at different times, it plays a pivotal role in comprehending and responding to dynamic changes on Earth. Therefore, its significance extends across various applications, including disaster monitoring, urban planning, vegetation change, and environmental monitoring [1], [2], [3], [4], [5], [6].

Remote sensing change detection primarily aims to identify semantic changes in temporal data, with temporal interaction playing a crucial role. Traditional methods often rely on concatenating or subtracting feature maps between bitemporal images in the spatial domain [7], [8], [9], [10]. However, this pixelwise interaction may not adequately capture changes in complex scenes, as illustrated in Fig. 1. In such cases, objects may be obscured by the surrounding background. In contrast, analyzing changes in the frequency domain offers a more nuanced approach, distinguishing between different regions associated with objects [11]. Low-frequency components capture global geometric structures and are useful for detecting overall changes, while high-frequency components represent detailed local information, such as edges, enhancing the precision of change boundaries. Therefore, it is always desirable to balance different components to simultaneously detect the overall changes while preserving the boundaries. Moreover, specific objects may favor different frequencies; for instance, textured objects benefit more from high-frequency components. Spatial-domain interactions lack the flexibility to adapt to such diverse scenarios. It should be noted that, noises may also generate high-frequency responses in the frequency domain, thereby requiring calibration between different components.

One of the challenges in remote sensing change detection is managing appearance shifts between bitemporal images. These shifts can arise from seasonal variations, atmospheric conditions, and other environmental factors, which often lead to pseudochanges. For instance, as shown in Fig. 1, the same road appears with different colors and brightness levels in the T1 and T2 images. To address this issue, domain adaptation (DA) techniques have been applied to minimize discrepancies between the source and target domains [12], [13], [14]. However, the existing methods typically focus on DA at the image level, overlooking the fact that change detection tasks involve both changed and unchanged regions. This approach can result in misalignments, decreased discriminative ability, and unexpected learning challenges, particularly in the presence of changed regions. A direct approach to mitigating these

Manuscript received 11 May 2024; revised 18 June 2024 and 30 June 2024; accepted 19 July 2024. Date of publication 23 July 2024; date of current version 1 August 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62371237 and Grant 62071421; in part by the State Key Laboratory of Geo-Information Engineering and Key Laboratory of Surveying and Mapping Science and Geospatial Information Technology of MNR, CASM; and in part by the Open Fund of Laboratory of Intelligent Control. *(Corresponding author: Fengchao Xiong.)*

Fengchao Xiong, Tianhan Li, and Jianfeng Lu are with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: fcxiong@njust.edu.cn).

Yi Yang is with the Research Center for Natural Resource Surveying and Monitoring, Chinese Academy of Surveying and Mapping, Beijing 100830, China.

Jun Zhou is with the School of Information and Communication Technology, Griffith University, Nathan, QLD 4111, Australia.

Yuntao Qian is with the College of Computer Science, Zhejiang University, Hangzhou 310027, China.

Digital Object Identifier 10.1109/TGRS.2024.3432819

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:24:32 UTC from IEEE Xplore. Restrictions apply.

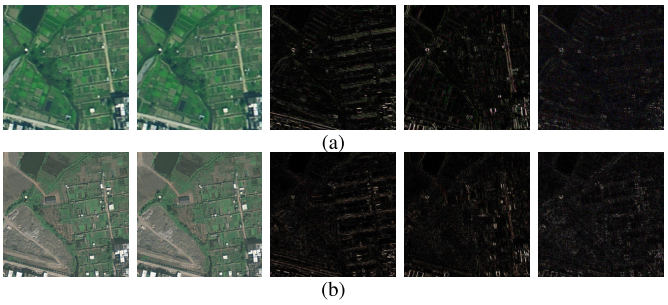


Fig. 1. Visual sample of CLCD dataset: (a) T1 and (b) T2 images. From left to right are the original images, and LL, LH, HL, and HH components, where "LL" corresponds to the low-frequency component, "LH" represents the vertical high-frequency component, "HL" is the horizontal high-frequency component, and "HH" is the high-frequency component in both vertical and horizontal directions (for visualization, LL, LH, HL, and HH components are shown with the same resolution).

issues could involve implementing DA in regions outside the annotated changed areas. Nevertheless, this strategy presents its own set of difficulties, as the only areas annotated for change are those of interest, and the remaining regions may also experience changes but are considered unimportant for the task at hand. This complicates the adaptation process and necessitates a more nuanced approach to effectively manage domain discrepancies.

Aiming at the aforementioned problems, we propose a wavelet Siamese network with semi-supervised DA, named WS-Net++, for remote sensing image change detection. WS-Net++ employs the wavelet-based spatial–frequency interaction module to extract feature differences. The spatial-domain interaction performs feature subtraction to capture the pixelwise differences. The frequency-domain interaction separates feature maps into low-frequency and high-frequency components and adaptively adjusts their contributions based on the image context, enhancing the flexible adaptation to different scenarios. The within- and between-frequency interactions are built to facilitate finding the complete changes. As a result, the network has more capacity to adaptively handle diverse scenarios with overall and subtle changes. During training, WS-Net++ divides all the regions into three groups, i.e., changed regions identified by the given labels and unchanged regions and regions of no interest through clustering. WS-Net++ then introduces a contrastive DA loss, which enlarges the feature distributions between changed regions while shortening the feature distributions between unchanged regions. Consequently, the semantic gap between bitemporal images is greatly reduced with enhanced discriminative ability, helping to tackle the pseudochanges. Extensive experiments demonstrate the effectiveness and superiority of our approach over competing methods.

In summary, the contributions of this article are threefold.

1) We introduce a novel wavelet-based spatial–frequency interaction module that simultaneously captures pixelwise interactions in the spatial domain and within-/between-frequency interactions in the frequency domain. This dual-domain interaction facilitates identifying the targeted changes with improved completeness of change boundaries.
2) We introduce the semi-supervised DA to explicitly narrow the domain discrepancy in the unchanged regions and widen the domain discrepancy between the changed regions. This contrastive adaptation not only produces discriminative domain-invariant representation but also mitigates the influence of regions with no interest.
3) Our WS-Net++ achieves the state-of-the-art performance on the LEVIR-CD [8], WHU-CD [15], and CLCD [16] datasets, achieving the $F1$ scores of 91.31%, 94.52%, and 79.77%, respectively.

A preliminary conference version of this work was presented in IGARSS 2023 [17]. In this extended version, we introduce a novel semi-supervised DA to address the appearance drift between bitemporal images. This enhancement yields a gain of **0.3%**, **1.14%**, and **1.39%** in $F1$ score across three datasets. Moreover, more comprehensive comparisons of different datasets, analysis of network components, and visual results are provided to offer a comprehensive understanding of the advantages and advancements of our work.

The remainder of this article is organized as follows. Section II discusses the related work. Section III introduces the details of the proposed network. Experimental results and analysis are provided in Section IV. Section V summarizes our work.

II. Related Works

This section presents recent advance of remote sensing image change detection, frequency-domain analysis in deep learning, and DA.

A. Change Detection

Traditional change detection methods can be categorized into four main approaches: algebra-based, transform-based, classification-based, and machine learning-based. Algebra-based methods, such as image differencing, image regression, and change vector analysis, focus on change detection but are sensitive to threshold selection [18], [19], [20]. Transform-based methods utilize data transformations, such as principal component analysis (PCA) [21], graph spectral analysis [22], and multivariate alteration detection [23], to emphasize changes. These methods often rely on clustering algorithms or Markov random fields (MRFs) for segmentation, which can present challenges with parameter selection. Classification-based methods identify changes according to precomputed classes [24] but may accumulate errors in the process. Machine learning-based approaches, such as those using the support vector machine [25], can struggle with robustness in complex change detection scenarios.

In recent years, a shift from traditional methods to deep-learning-based methods has occurred due to their superior discriminative feature extraction capabilities and the increasing availability of remote sensing images [6], [26], [27]. Most deep learning methods [10], [28], [29] employ a Siamese network structure and are categorized into early and late fusion approaches. Late fusion methods typically outperform

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:24:32 UTC from IEEE Xplore. Restrictions apply.

early fusion by highlighting differences between images more effectively and achieving superior performance [7].

Significant efforts have been made to design robust representation networks to highlight differences between temporal images [9], [10], [30], [31], [32]. The U-Net architecture, known for its multiscale feature extraction capabilities, is commonly utilized in this context [33], [34]. To enhance feature representation further, attention mechanisms are frequently integrated, directing the network to focus on critical areas [35], [36]. Various forms of attention, including spatial attention [8], [37], channel attention [38], [39], self-attention [9], [10], and cross attention [40], have been explored to augment change detection capabilities. For instance, Chen and Shi [8] developed STANet, leveraging self-attention to capture spatial–temporal dependencies. Feng et al. [29], building on the concept of cross attention, introduced an intrascale cross-interaction module that leverages the complementarity between fine-grained local details and global interests. Similarly, Tang et al. [41] designed a deformable attention Transformer, combining deformable convolution and multihead self-attention to capture global context dependencies in remote sensing images. Feng et al. [26] introduced an intertemporal joint-attention block that unifies self-attention and cross attention to suppress task-irrelevant interferences while focusing on genuine changes. In addition, Bai et al. [31] integrated edge structure prior into CNN for change prediction, resulting in promising performance. Moreover, leveraging the powerful representation of the foundation model, the performance is further improved [42], [43].

However, relying solely on designing different network architectures to enhance representation for defense against pseudochanges may lack reliability. These architectures still struggle to ensure invariant feature extraction between the same targets captured at different times. Therefore, this article adopts a semi-supervised DA approach to align the feature distribution between bitemporal images, addressing this challenge effectively.

In addition to feature extraction, modeling temporal interactions for difference extraction is crucial in change detection. Common approaches involve feature concatenation or subtraction. To capture nonlinear correlations between bitemporal images, some methods, such as [31] and [44], have adopted recurrent neural networks (RNNs) for change detection. Furthermore, techniques, such as spatialwise and channelwise attention, have been utilized to refine feature changes in [45], [46], [47], and [48]. However, these methods perform temporal interaction solely in the spatial domain and can hardly enjoy the discriminative clues in the frequency-domain clues. Tang et al. [49] applied different interaction methods based on the characteristics of different frequency components. Nonetheless, their approach assumes equal contributions from different frequency components, ignoring that some components related to noise may provide harmful clues for change detection, and the interaction between different frequency components is neglected. Therefore, we introduce a learnable weight to adaptively adjust the importance of different components and promote cross-frequency interaction.

B. Frequency-Domain Analysis in Deep Learning

Signals can be converted from the time and space domain into the frequency domain using designed mathematical functions, such as discrete cosine transform (DCT) [50], Fourier transform [51], and wavelet transform [52], [53]. Frequency-domain analysis can provide complementary insights to identify signal characteristics. As a result, it has been utilized in various tasks, such as face forgery detection [54], pansharpening [55], and change detection [51], [56], [57], [58]. For example, Dräger et al. [59] injected the trigger image into the poisoned image in the low-frequency components produced by 2-D discrete wavelet transformation to achieve stealthy data poisoning. In [60], the multilevel wavelet transformation was adopted to retain high-frequency information for super-resolution. Zhou et al. [55] selected desired frequencies from frequency features obtained via DCT and combined them with spatial-domain features for change detection. Xue et al. [57] applied DCT to feature maps and performed global average pooling on each frequency to model channel correlation for enhanced representation. Yuan et al. [51] performed convolution in the Fourier domain to better preserve edge information in the change region. However, most of these methods focus on enhancing network representation with frequency-domain operations. In contrast, we aim to establish an interaction between bitemporal images in both spatial and frequency domains.

C. Domain Adaption

DA aims to enable a model trained on a source domain to perform well in the target domain. Depending on the availability of labels, DA can be categorized into unsupervised, semi-supervised, and supervised approaches [61]. In unsupervised DA, only labels in the source domain are available. Semi-supervised DA assumes that there are some labeled data in the target domain in addition to the labeled source data. Supervised DA deals with scenarios where both labeled source and target data are accessible. The main task of DA is to formally reduce the domain shift in data distribution of the source and target domains. As reviewed in [62], discrepancy-based, adversarial-based, and reconstruction-based are the main approaches. The discrepancy-based approach leverages different criteria, such as statistical criteria, to minimize the shift between two domains. For instance, maximum mean discrepancy was regarded as a measure and employed in [63] to learn domain-invariant feature representations between the source and target domains. Adversarial-based approaches encourage domain confusion, so that the discriminative network cannot distinguish between the source and target domains. Ganin and Lempitsky [64] augmented the standard architecture with a gradient reversal layer (GRL) to maximize the domain confusion loss, promoting feature adaptation in both domains. Reconstruction-based approaches treat the reconstruction of source or target samples as an auxiliary task to learn shared encoding between the two domains [65].

When it comes to change detection, each pixel in the image can be considered a sample. The DA task involves aligning the distribution of pixels in bitemporal image pairs

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:24:32 UTC from IEEE Xplore. Restrictions apply.

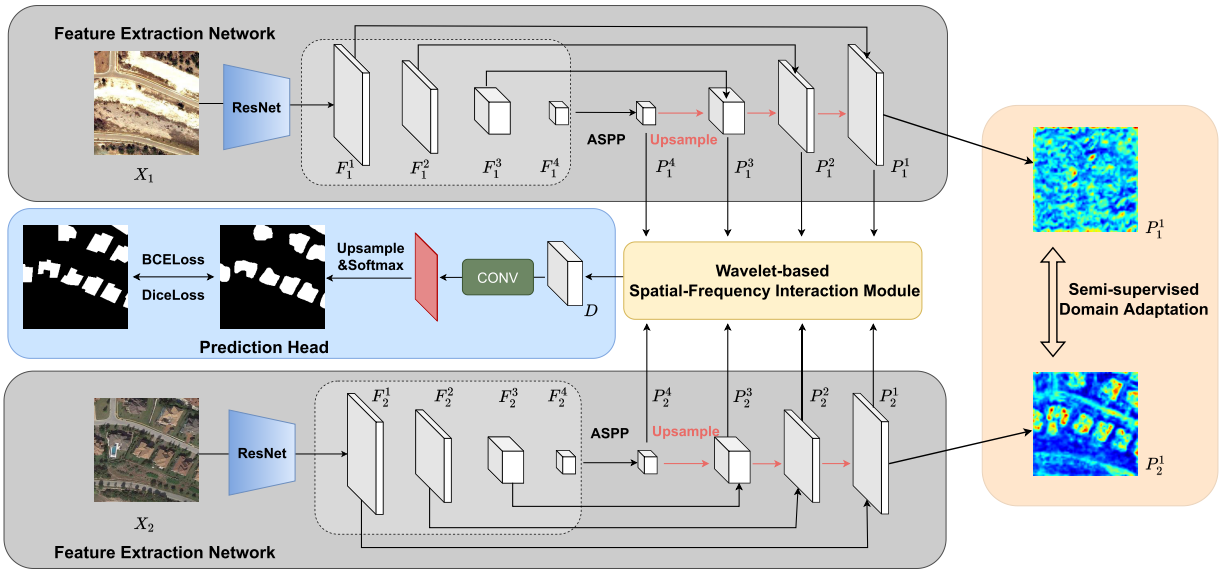


Fig. 2. Illustration of the proposed WS-Net++ where "F" and "P" denotes the extracted feature maps with the ResNet and FPN, respectively. The network comprises a feature extraction network, a wavelet-based spatial–frequency interaction module, and a prediction head. The wavelet-based spatial–frequency interaction module extracts semantic differences between bitemporal images. DA bridges the semantic gap between bitemporal images with enhanced discriminative ability.

to mitigate domain shift caused by luminance fluctuations, seasonal changes, and more. Liu et al. [12] leveraged generative adversarial learning to perform image translation for appearance alignment and feature adaptation for distribution alignment, suppressing pseudochanges. Zhao et al. [13] minimized the distance of the Gram matrix of bitemporal images for DA. In addition, Zhang et al. [66] reduced the maximum mean discrepancy to align heterogeneous optical and SAR images. Unfortunately, all these methods minimize the domain discrepancy but overlook that the image content in bitemporal image pairs always differs in change detection tasks. Some regions, such as the changed ones, belong to different classes, which may lead to misalignment and poor performance. This article categorizes the image into changed, unchanged, and irrelevant regions and employs the semi-supervised DA to learn discriminative domain-invariant representations.

III. Proposed Method

This section presents the details of our method, including the overall network, wavelet-based spatial–frequency interaction module, and semi-supervised DA.

A. Network Overview

Fig. 2 illustrates the framework of the proposed WS-Net++. WS-Net++ is structured as a Siamese network, consisting of a feature extraction network, a wavelet-based spatial–frequency interaction (WSFI) module, and a prediction head. The feature extraction module incorporates a ResNet-18 architecture alongside a feature pyramid network (FPN) [67] to extract multiscale features $\{P_1^i, P_2^i\}$ from bitemporal images. The index $i \in \{1, 2, 3, 4\}$ represents the different scales of the extracted features. To expand the receptive field and enhance feature representation, the atrous spatial pyramid pooling (ASPP) module [68] is applied to the two feature maps with the smallest spatial resolution. The WSFI module builds the spatial–frequency correlation between bitemporal images for subsequent change detection. The prediction head employs a simple multilayer perceptron (MLP) to calculate the probability of each pixel undergoing a change, resulting in the final binary map of the change. The probability calculation is formulated as follows:

$$p = \text{softmax}(\text{Up}(\text{conv}_{1\times1}(D))). \tag{1}$$

Here, D represents the concatenation of feature differences at different scales and $\text{Up}(\cdot)$ is the upsampling operation. To minimize the appearance discrepancy between bitemporal images, we apply semi-supervised DA. In Sections III-B and III-C, we provide detailed explanations of the WSFI module and the DA process.

B. Wavelet-Based Spatial–Frequency Interaction Module

As depicted in Fig. 3, our wavelet-based spatial–frequency interaction computes feature differences in both spatial and frequency domains. The spatial-domain interaction focuses on pixel-level differences by subtracting feature maps between P_1^i and P_2^i, which is expressed as follows:

$$D_s^i = \text{conv}_{3\times3}\left(\left|P_1^i - P_2^i\right|\right) \tag{2}$$

where $\text{conv}_{3\times3}$ denotes a 3×3 convolution operation, followed by batch normalization and a rectified linear unit for nonlinear transformation. The symbol $|\cdot|$ represents the absolute value operation.

Change detection can be viewed as a dense prediction task where both overall structure and boundary precision are important. The overall structure is associated with low-frequency components, while boundary precision is concerned with high-frequency components. In addition, different objects may prefer different frequency components for effective change

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:24:32 UTC from IEEE Xplore. Restrictions apply.

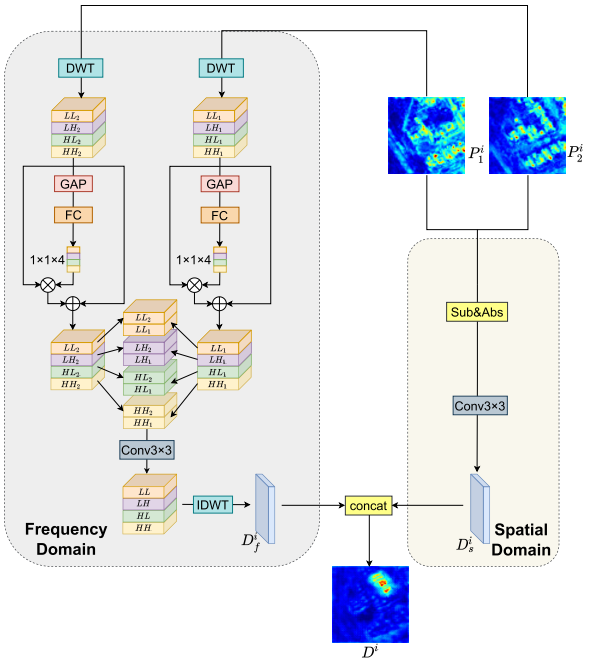


Fig. 3. Illustration of the wavelet-based spatial–frequency interaction.

detection. Spatial-domain interaction lacks the flexibility and capability to adequately address both overall change detection and the completeness of change boundaries. Consequently, this can lead to patterns that either miss overall changes or present incomplete boundaries. To address these challenges, the frequency-domain interaction is also integrated into the WSFI module.

Regarding frequency-domain interaction, we first apply the discrete wavelet transform (DWT) to decompose the feature map P into four distinct frequency maps, represented as $\{P_{\mathrm{LL}}, P_{\mathrm{LH}}, P_{\mathrm{HL}}, P_{\mathrm{HH}}\}$. Using the 2-D Haar DWT, these frequency maps are generated by performing the following convolutions:

$$\begin{aligned} P_{\mathrm{LL}} &= P \otimes \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \downarrow \\ P_{\mathrm{LH}} &= P \otimes \begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix} \downarrow \\ P_{\mathrm{HL}} &= P \otimes \begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix} \downarrow \\ P_{\mathrm{HH}} &= P \otimes \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \downarrow \end{aligned} \tag{3}$$

where $\downarrow$ denotes the downsampling operator. Among the decomposed frequency maps, P_{LL} corresponds to the low-frequency component, P_{LH} to the vertical high-frequency component, P_{HL} to the horizontal high-frequency component, and P_{HH} to the high-frequency component in both vertical and horizontal directions. To selectively focus on the critical frequency components for effective change detection, we introduce learnable enhancement to adjust the contributions of each component. We measure their contributions $w = [w_{\mathrm{LL}}, w_{\mathrm{LH}}, w_{\mathrm{HL}}, w_{\mathrm{HH}}] \in \mathbb{R}^{1\times 4}$ based on the image context, i.e.,

$$w = \tanh(\mathrm{relu}(\mathrm{FC}(\mathrm{GAP}(\mathrm{concat}(P_{\mathrm{LL}}, P_{\mathrm{LH}}, P_{\mathrm{HL}}, P_{\mathrm{HH}}))))). \tag{4}$$

Here, concat$(\cdot)$ denotes feature concatenation along the channels, GAP$(\cdot)$ represents the global average pooling process, and FC$(\cdot)$ refers to a fully connected layer that reduces the dimensionality to four. Following these operations, a residual connection is employed. For instance, the modulation of the low-frequency component P_{LL} is implemented as follows:

$$P'_{\mathrm{LL}} = P_{\mathrm{LL}} \times w_{\mathrm{LL}} + P_{\mathrm{LL}}. \tag{5}$$

In this way, the network has more adaptability to complex change detection scenarios.

The bitemporal interaction is performed between corresponding frequency components. Specifically, low-frequency interaction is achieved through feature concatenation within the low-frequency components, while high-frequency interaction follows the same principle. The resultant interaction is then convolved with a 3×3 convolution to facilitate cross-frequency interactions, thereby harnessing complementary information across different frequency components for higher detection accuracy. Following the interaction among various frequency components, the feature change map is transformed back to the spatial domain using the inverse discrete wavelet transform (IDWT), yielding the frequency-domain interaction D_f^i.

The spatial-domain interaction and the frequency-domain interaction are concatenated to yield the final interaction D_i. To promote multiscale interaction, the values of ${D^i}_{i=1,2,3,4}$ are densely integrated as follows:

$$D^i = \mathrm{concat}(\mathrm{Up}([D^{i+1}, \ldots, D^4]), D^i). \tag{6}$$

Here, Up$(\cdot)$ denotes upsampling followed by a 1×1 convolution, which aligns the dimensionality of the feature maps across different scales.

C. Semi-Supervised Domain Adaptation

There is an obvious appearance disparity between bitemporal images, resulting from imaging conditions, seasons, and more, as clearly shown in Fig. 4. Essentially, these interrupting factors give rise to feature domain mismatches for the same object and can easily mislead a network into producing pseudochanges. To alleviate this issue, DA is required to align the feature maps of the same object in the same space. However, the challenge of DA for change detection arises when only the labels of changed objects are available. One naive approach is to consider all the remaining regions as unchanged. Due to the labeling scheme, not all the remaining regions contain no change. Some changes have occurred but are considered irrelevant, which may introduce negative DA.

The proposed WS-Net++ incorporates DA in a semi-supervised manner. As shown in Fig. 4, we partition the bitemporal images into three distinct regions: changed regions, unchanged regions, and regions of no interest. Given the nature of the change detection task, there are inherent differences between the changed areas. Leveraging this insight, we obtain

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:24:32 UTC from IEEE Xplore. Restrictions apply.

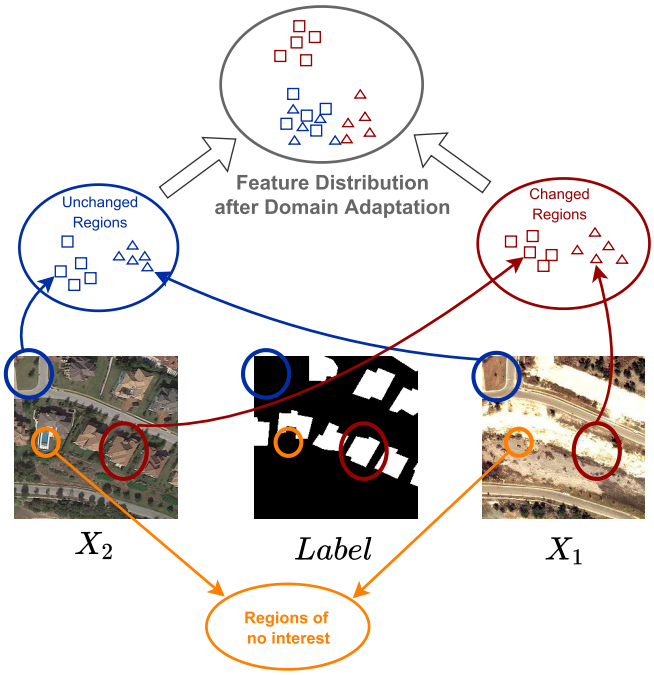


Fig. 4. Schematic of our semi-supervised domain adaptation.

the changed regions in a supervised manner, identified using ground-truth labels from the provided change map. It is worth noting that, as the feature maps undergo downsampling, they end up with different spatial resolutions compared with the ground-truth change map. One approach is to downsample the change map to match the spatial resolution of the feature map. However, with multiple downsampling steps, achieving pixelwise consistency between the downscaled change map and the feature map becomes challenging. Therefore, DA is exclusively applied to the feature map with the highest resolution, i.e., $\{P_1, P_2\}$.

The remaining two regions are obtained in an unsupervised manner, specifically through clustering. Given two feature maps $\{P_1, P_2\}$, we first measure the pixelwise L_2 distances in the remaining pixels, generating the following distance map M:

$$M_k = \left\| P_1^k - P_2^k \right\|_2 \tag{7}$$

where k indexes the position. Subsequently, we apply the K-means algorithm to M, resulting in two clusters, and calculate the average L_2 distance of all the pixels in the cluster. In general, the unchanged regions exhibit more similarities than regions of no interest, leading to a lower average distance. Based on this assumption, the unchanged regions and regions of no interest are identified.

The DA includes two parts. The first part is the intraclass domain discrepancy loss, which compacts the feature representations within the unchanged regions. The second part is the interclass domain discrepancy loss, which enlarges the feature representations between the changed regions. The intraclass and interclass discrepancies are contrastively optimized to enhance the adaptation performance. Mathematically, the DA loss is defined as follows:

$$\mathcal{L}_{\text{dom}} = \frac{1}{N_{\text{un}}} \sum_{k\in\{\text{Unchanged}\}} \exp\left(\frac{\|P_1^k - P_2^k\|_2}{\tau}\right) + \frac{1}{N_{\text{ch}}} \sum_{k\in\{\text{Changed}\}} \exp\left(-\frac{\|P_1^k - P_2^k\|_2}{\tau}\right). \tag{8}$$

Here, N_{ch} and N_{un} are the number of pixels in the changed set and unchanged set, respectively. The τ is the temperature parameter and is set to 1 in our implementation. There are also cases where all the regions in a mini-batch undergo no changes. In such instances, the discriminative loss part is eliminated.

Incorporating the DA loss, the cost function of our WS-Net++ is a multitask loss defined as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}} + \lambda \mathcal{L}_{\text{dom}}. \tag{9}$$

Here, $\mathcal{L}_{\text{BCE}}$ is the classification loss, while $\mathcal{L}_{\text{Dice}}$ aims to mitigate the sample imbalance problem between changed and unchanged areas. Their definitions are given by

$$\mathcal{L}_{\text{BCE}} = -\sum_{i=1}^{N} t_i \log p_i + (1 - t_i) \log(1 - p_i)$$
$$\mathcal{L}_{\text{Dice}} = \sum_{i=1}^{N} 1 - \frac{2 p_i t_i}{p_i + t_i}. \tag{10}$$

Here, p_i represents the predicted probability value output by the network, and t_i and $1 - t_i$ correspond to the ground-truth labels.

Through experiments, we experimentally set λ to 0.01 to balance each loss function. We will examine its impact on the experiment section. In our training process, the K-means clustering is only employed in the first ten training epochs. The main reason is that we found that with the training process proceeds, the discriminative ability of the network is gradually enhanced, more focusing on the target of interest while ignoring the rest targets. As a result, the regions of no interest were easily recognizable without the need for K-means clustering, which can be a time-consuming process.

IV. Experiments

This section compares our WS-Net++ with alternative methods to demonstrate its efficacy in remote sensing change detection. A detailed ablation study is provided to analyze the behavior and effectiveness of different components.

A. Experimental Settings

1) Datasets: We employed three datasets for evaluation: LEVIR-CD, WHU-CD, and CLCD. The LEVIR-CD dataset [8] is dedicated to building change detection and comprises 637 pairs of bitemporal images with a size of 1024 × 1024 pixels. As in [8], each image is segmented into nonoverlapping small patches of 256 × 256 pixels, resulting in a total of 7120/1024/2048 samples distributed across training, validation, and testing phases, respectively.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:24:32 UTC from IEEE Xplore. Restrictions apply.

The WHU-CD dataset [15] consists of a single pair of aerial images with 32 507 × 15 354 pixels, capturing 12 796 buildings in 2012 and 16 077 buildings in 2016. This image pair is also divided into nonoverlapping patches of 256 × 256 pixels. Subsequently, the dataset is randomly split into 6096/762/762 samples for training, validation, and testing, respectively.

Finally, the CLCD dataset [16] is a cropland change dataset comprising 600 pairs of images, each measuring 512 × 512 pixels. This dataset captures various changes in croplands, including alterations in buildings, roads, lakes, and bare soil lands. Consistent with the protocol in [16], the dataset is segmented into 360/120/120 samples for the training, validation, and testing stages, respectively.

2) Evaluation Metrics: The performance was measured with five metrics: precision (P), recall (R), $F1$ score ($F1$), intersection over union (IoU), and overall accuracy (OA). Below is the definition of these metrics

$$\begin{aligned} P &= \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \\ R &= \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \\ \mathrm{IoU} &= \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}+\mathrm{FP}} \\ \mathrm{OA} &= \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FN}+\mathrm{FP}} \\ F1 &= \frac{2}{R^{-1}+P^{-1}} \end{aligned} \tag{11}$$

where TP, FP, TN, and FN represent the number of true positive, false positive, true negative, and false negative, respectively. In general, the $F1$ index is usually used for ranking the performance.

3) Implementation Details: Our method was implemented using the PyTorch framework and trained on NVIDIA GeForce RTX 3090 GPUs, utilizing the AdamW optimizer. The training process commenced with an initial learning rate of 0.002 and modulated using the OneCycleLR method with a maximum of 0.002 and a minimum of 0.002/500. Specifically, for the LEVIR-CD dataset, we set the learning rate to 0.002, employed a batch size of 32, and conducted training for 250 epochs. For the WHU-CD dataset, the learning rate was set to 0.001, maintaining the same batch size of 32 over 250 epochs. In the case of the CLCD dataset, the learning rate was adjusted to 0.001, the batch size was reduced to 8, and the number of training epochs was extended to 350.

4) Compared Methods: We included a total of 14 methods for comparison, covering both CNN and Transformer-based approaches, as well as their combinations. The CNN-based methods include FC-EF, FC-Siam-Diff, FC-Siam-Conc [7], STANet [8], DTCDSCN [39], EGRCNN [31], DMINet [26], WS-Net [17], HFA-Net [50], and SDACD [12]. SDACD achieved DA between bitemporal images with image adaptation and feature adaptation. Transformer-based methods consist of BIT [9] and ChangeFormer [10]. Hybrid approaches that combine CNN and Transformer architectures include ICIF-Net [29] and WNet [41]. All performance metrics are referenced from the reported values in the original papers. In cases where this information was not available, we utilized publicly available code and default parameters for training and evaluation.

TABLE I

COMPARISON OF DIFFERENT METHODS ON RESULTS ON THE LEVIR-CD DATASET (**BOLD**: BEST AND UNDERLINE: SECOND BEST)

Method	P	R	F1	IoU	OA
FC-EF [7]	86.91	80.17	83.40	71.53	98.39
FC-Siam-Diff [7]	89.53	83.31	86.31	75.92	98.67
FC-Siam-Conc [7]	91.99	76.77	83.69	71.96	98.49
STANet [8]	83.81	<u>91.00</u>	87.26	77.40	98.66
DTCDSCN [39]	88.53	86.83	87.67	78.05	98.77
EGRCNN [31]	88.58	**91.72**	90.12	82.03	98.97
ChangeFormer [10]	92.05	88.80	90.40	82.48	99.04
BIT [9]	89.24	89.37	89.31	80.68	98.92
ICIF-Net [29]	91.39	89.24	90.38	82.31	98.99
DMINet [26]	92.52	88.86	90.70	82.99	99.07
WNet [41]	91.23	89.62	90.42	82.51	99.03
SDACD [12]	91.21	90.87	<u>91.04</u>	<u>83.58</u>	99.09
HFA-Net [50]	90.92	86.28	88.54	79.44	98.86
WS-Net [17]	**92.78**	89.31	91.01	83.50	<u>99.10</u>
WS-Net++(**Ours**)	<u>92.57</u>	90.08	**91.31**	**84.01**	**99.13**

B. Comparisons With State of the Arts

1) Results on the LEVIR-CD Dataset: Table I presents a comprehensive comparison of various methods on the LEVIR-CD dataset. WS-Net++ stands out as the top performer in precision, $F1$, and IoU, demonstrating its effectiveness in change detection tasks. In contrast to CNN-based approaches, Transformer-related methods generally exhibit superior performance owing to their robust feature extraction capabilities. Notably, with ResNet-18 as the backbone network, both WS-Net and WS-Net++ outperform several Transformer-related methods to a noticeable margin, as evidenced by their higher $F1$ scores. This observation underscores the effectiveness of our wavelet-based feature spatial–frequency interaction module, which translates feature maps into the frequency domain and dynamically guides the network to pay attention to the critical components. By this means, the overall structural and local detailed changes can be more adequately handled for enhanced change detection performance. For this reason, our method also surpasses the HFA-Net a noticeable margin. Furthermore, the incorporation of DA enforces the feature consistency between the bitemporal images with augmented discriminative ability, substantially reducing the risk of false detections and misdetections. Consequently, WS-Net++ surpasses WS-Net, achieving a noteworthy gain of **0.3%** in $F1$. SDACD globally aligns bitemporal images but carries the risk of negative alignment. In contrast, WS-Net++ achieves domain alignment by simultaneously minimizing the intraclass domain discrepancy in the unchanged regions and maximizing the interclass domain discrepancy in the changed regions. Consequently, WS-Net++ outperforms SDACD.

In order to examine the impact of λ on the change detection performance, we varied λ within the range 0.001, 0.005, 0.01, 0.05, and 0.1 in Table II. As can be observed, the best performance is achieved at $\lambda = 0.01$. Therefore, $\lambda = 0.01$ is set in our experiment.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:24:32 UTC from IEEE Xplore. Restrictions apply.

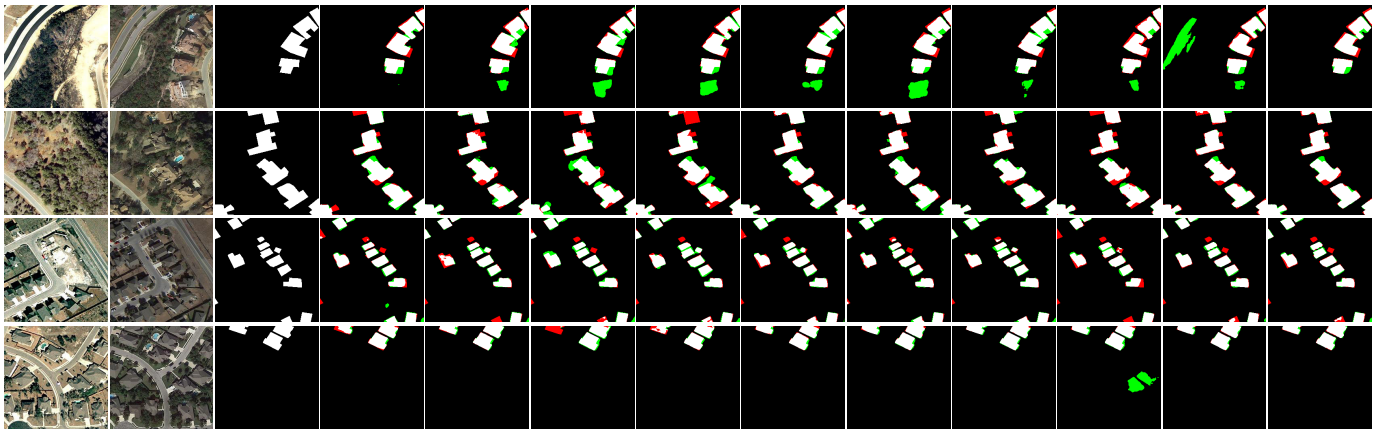


Fig. 5. Visual comparison on the LEVIR-CD dataset. (a) T1 images. (b) T2 images. (c) Ground truth. (d) BIT. (e) ChangeFormer. (f) EGRCNN. (g) ICIF-Net. (h) DMINet. (i) WNet. (j) SDACD. (k) HFA-Net. (l) WS-Net. (m) Ours.

TABLE II
IMPACT λ ON THE PERFORMANCE

Weight	P	R	F1	IoU	OA
0.1	**93.05**	89.52	91.27	83.92	99.12
0.05	92.96	89.60	91.25	83.91	99.12
0.01	92.57	**90.08**	**91.31**	**84.01**	**99.13**
0.005	92.86	89.51	91.16	83.75	99.11
0.001	92.13	90.03	91.07	83.61	99.10

We provide a qualitative comparison on the LEVIR-CD dataset in Fig. 5. In the figure, true positives and true negatives are denoted by white and black, while false positives and false negatives are indicated by green and red. Ideally, a high-quality change map is expected to exhibit fewer red and green colors. Here, FC-EF, FC-Siam-Diff, FC-Siam-Conc, STANet, and DTCDSCN are excluded due to their limited effectiveness. With less red and green colors, our WS-Net++ matches well with the ground-truth change map, indicating its exceptional performance. Our WS-Net++ outperforms BIT, ChangeFormer, and EGRCNN with more accurate edges because of our wavelet-based spatial–frequency feature difference extraction, which can adaptively adjust the contribution of low- and high-frequency features to detect subtle changes. As observed in the first row, significant differences in imaging conditions lead to appearance drift between images. The ruins in the background are misidentified as buildings by many other methods, leading to false positives. Without proper alignment between bitemporal images, other methods exhibit noticeable false negatives, indicating misdetection. In contrast, the feature alignment achieved by semi-supervised DA enables WS-Net++ to more accurately distinguish between changed and unchanged areas, mitigating the impact of appearance drift and enhancing overall detection accuracy.

2) Results on the WHU-CD Dataset: Table III provides a comprehensive comparison of all methods on the WHU-CD dataset. Due to the spatial–frequency interaction facilitated by the wavelet-based approach and the feature alignment achieved through DA between bitemporal images, WS-Net++ outperforms alternative methods across all indexes. This notable achievement in quantitative performance is further supported by the visual comparison in Fig. 6, which demonstrates superior qualitative performance with fewer false positives and false negatives. Overall, the promising results on the WHU-CD

TABLE III
COMPARISON OF DIFFERENT METHODS ON THE WHU-CD DATASET (**BOLD**: BEST AND UNDERLINE: SECOND BEST)

Method	P	R	F1	IoU	OA
FC-EF [7]	83.54	73.85	78.39	64.47	98.21
FC-Siam-Diff [7]	85.92	78.89	82.26	69.86	98.50
FC-Siam-Conc [7]	82.46	85.24	83.83	72.16	98.55
STANet [8]	85.10	79.40	82.20	69.70	98.50
DTCDSCN [8]	91.42	87.60	89.47	80.94	99.09
EGRCNN [31]	92.60	86.55	89.47	80.04	99.10
ChangeFormer [10]	92.06	83.46	87.55	77.86	98.96
BIT [9]	93.91	87.84	90.78	83.11	99.21
ICIF-Net [29]	91.19	85.92	88.48	79.34	99.01
DMINet [26]	82.87	87.54	85.14	74.12	98.65
WNet [41]	94.17	83.94	88.76	79.79	99.06
FA-Net [50]	89.76	84.28	86.94	76.89	98.88
SDACD [12]	91.21	88.95	90.07	81.93	99.14
WS-Net [17]	**95.88**	91.01	93.38	87.58	99.43
WS-Net++(**Ours**)	95.36	**93.70**	**94.52**	**89.61**	**99.52**

TABLE IV
COMPARISON OF DIFFERENT METHODS ON THE CLCD DATASET (**BOLD**: BEST AND UNDERLINE: SECOND BEST)

Method	P	R	F1	IoU	OA
FC-EF [7]	58.88	56.32	57.57	40.42	93.82
FC-Siam-Diff [7]	59.27	62.38	60.79	43,66	94.31
FC-Siam-Conc [7]	61.71	65.29	63.00	45.99	94.85
STANet [8]	55.80	68.00	61.30	38.40	93.60
DTCDSCN [39]	61.25	59.11	60.16	43.02	94.18
EGRCNN [31]	67.12	69.04	68.07	51.59	95.18
ChangeFormer [10]	58.29	47.25	52.19	35.31	93.56
BIT [9]	64.18	58.63	61.28	44.18	94.48
ICIF-Net [29]	66.84	54.02	58.75	42.60	94.58
DMINet [26]	70.30	46.40	55.90	38.79	94.55
WNet [41]	68.45	57.82	62.69	45.66	94.88
SDACD [12]	64.80	65.87	65.38	48.56	94.80
HFA-Net [50]	72.15	57.05	63.72	46.75	95.17
WS-Net [17]	**84.85**	72.82	78.38	64.45	97.01
WS-Net++(**Ours**)	84.58	**75.47**	**79.77**	**66.34**	**97.15**

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:24:32 UTC from IEEE Xplore. Restrictions apply.

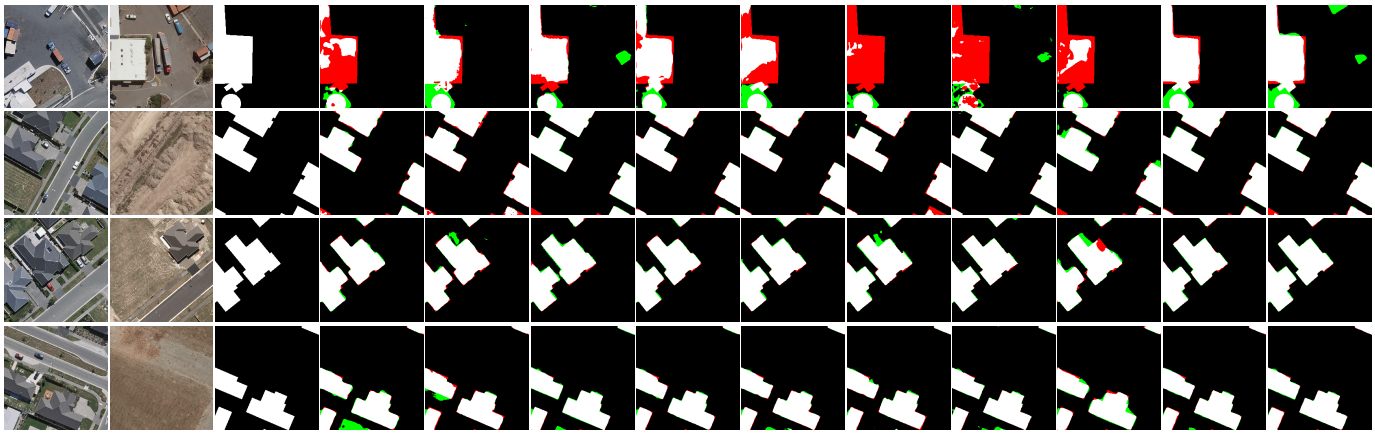


Fig. 6. Visual comparison on the WHU-CD dataset. (a) T1 images. (b) T2 images. (c) Ground truth. (d) BIT. (e) ChangeFormer. (f) EGRCNN. (g) ICIF-Net. (h) DMINet. (i) WNet. (j) SDACD. (k) HFA-Net. (l) WS-Net. (m) Ours.

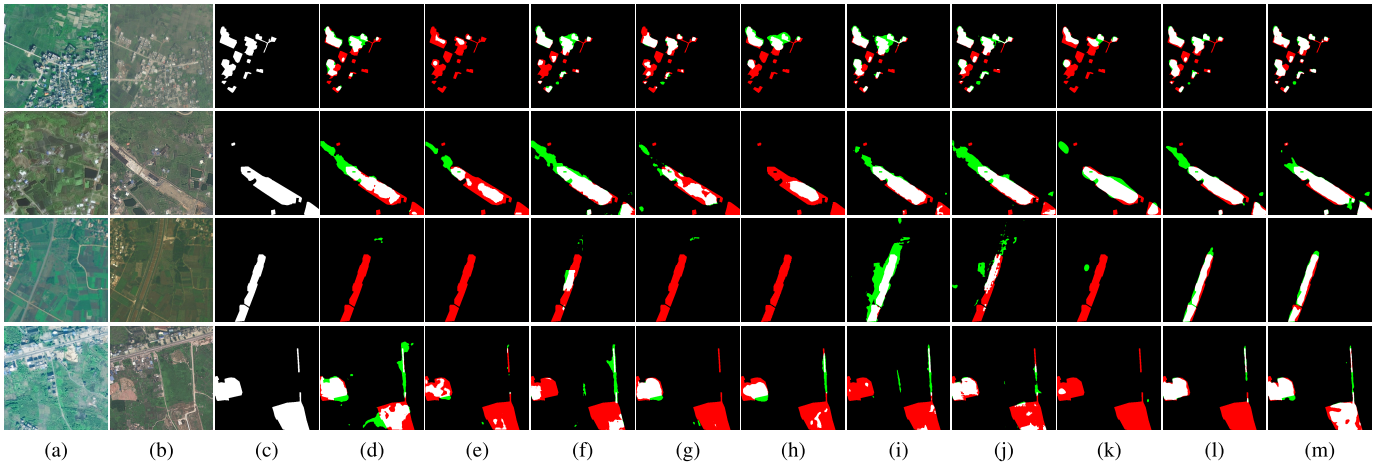


Fig. 7. Visual comparison on the CLCD dataset. (a) T1 images. (b) T2 images. (c) Ground truth. (d) BIT. (e) ChangeFormer. (f) EGRCNN. (g) ICIF-Net. (h) DMINet. (i) WNet. (j) SDACD. (k) HFA-Net. (l) WS-Net. (m) Ours.

dataset strongly underscore the effectiveness of our method in change detection, highlighting the advantages of the proposed spatial–frequency interaction and DA.

3) Results on the CLCD Dataset: The CLCD dataset encompasses numerous subtle changes related to croplands. Varying imaging conditions and the complex background environment raise the difficulty for change detection. Table IV reveals that many methods struggle to detect changes, resulting in lower performance. The wavelet-based spatial–frequency interaction allows for adaptive adjustment of contributions between details and structural interactions within bitemporal images. Consequently, WS-Net and WS-Net++ exhibit superior capabilities in detecting changes and achieve noticeable performance gains. The semi-supervised DA between bitemporal images further minimizes intraclass discrepancy and maximizes interclass margin, significantly contributing to the exceptional performance of WS-Net++ compared with all other methods.

The visual comparison is presented in Fig. 7. As observed from the last column, WS-Net++ demonstrates significantly higher efficacy compared with all other methods, showing its capability to address the complexities and diversities in complex scenarios. In the first scene, where subtle changes exist, methods relying solely on spatial-domain interaction struggle, leading to noticeable false negatives. The changes in the second and third scenes are primarily related to newly built roads, presenting challenges due to complex background conditions. Leveraging wavelet-based spatial–frequency interaction, WS-Net++ exhibits fewer false positives. In the last scene, changes are associated with the cultivation of cropland, and most methods are unable to detect these changes. Because of more accurate alignment between different images, WS-Net++ effectively detects these changes.

C. Ablation Study

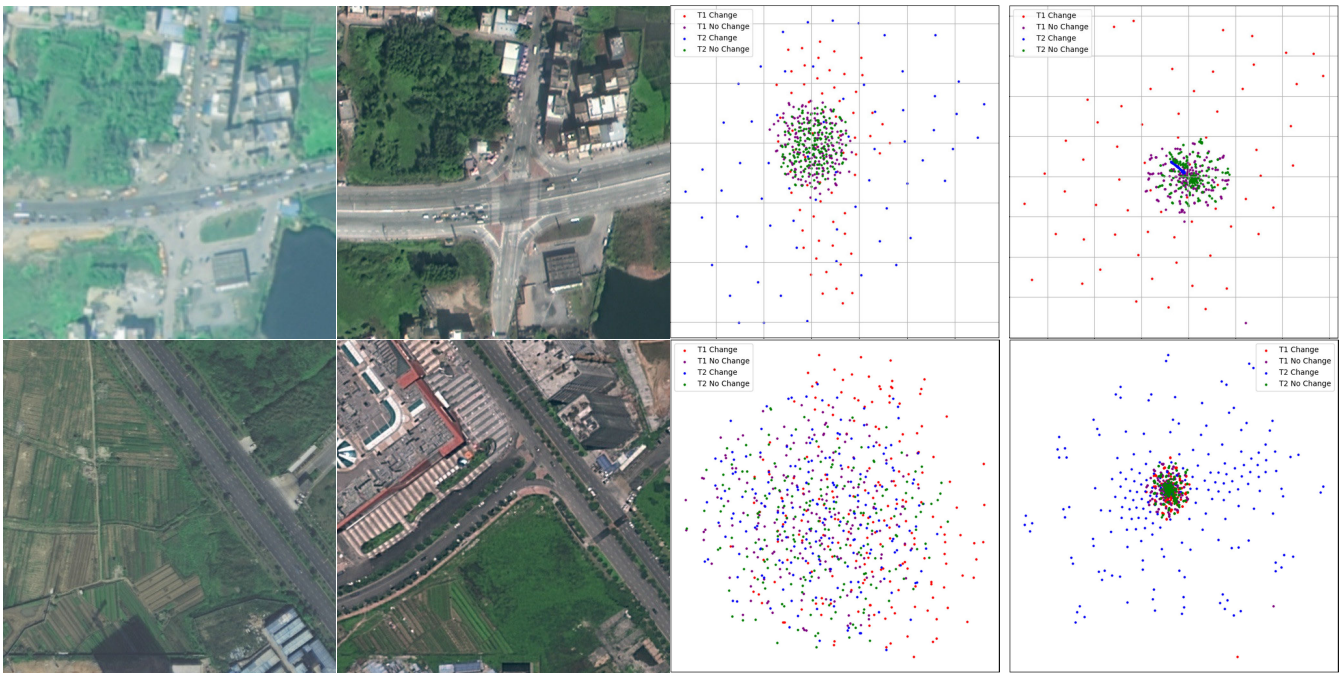
This section provides a comprehensive ablation study of each component in our WS-Net++.

1) Effectiveness of the Introduced Modules: We first evaluate the effectiveness of our wavelet-based spatial–frequency interaction and contrastive learning in Table V. The table reveals that all components contribute positively to the improved performance. Specifically, the wavelet-based spatial–frequency interaction is more capable of detecting the changes with higher completeness, contributing to the gains of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:24:32 UTC from IEEE Xplore. Restrictions apply.

TABLE V
ABLATION STUDY ON DIFFERENT MODULES

Baseline	WSFI	DA	LEVIR-CD P/R/F1/IoU/OA	WHU-CD P/R/F1/IoU/OA	CLCD P/R/F1/IoU/OA
✓	✗	✗	92.24/89.34/90.77/83.10/99.07	95.07/91.26/93.13/87.14/99.41	77.63/**76.39**/77.01/62.61/96.61
✓	✓	✗	**92.83**/89.25/91.01/83.50/99.10	**96.58**/91.86/94.16/88.97/99.50	84.34/74.29/79.00/65.28/97.06
✓	✗	✓	92.47/89.77/91.10/83.66/99.11	96.28/91.79/93.98/88.64/99.48	83.33/74.32/78.57/64.70/96.98
✓	✓	✓	92.57/**90.08/91.31/84.01/99.13**	95.36/**93.70/94.52/89.61/99.52**	**84.58**/75.47/**79.77/66.34/97.15**



(a) (b) (c) (d)

Fig. 8. Effectiveness of our semi-supervised domain adaptation. (a) T1. (b) T2. (c) Before. (d) After.

0.24%, **1.03%**, and **1.99%** in the $F1$ index. Semi-supervised DA can more accurately align the feature distribution between bitemporal images, leading to an improvement of **0.33%**, **0.85%**, and **1.56%** in the $F1$ index. The inclusion of two components allows for noticeable performance gain. This phenomenon evidently highlights the indispensable contribution of the introduced modules.

2) Effectiveness of Spatial and Frequency Interaction: Here, we study the impact of the wavelet-based spatial–frequency interaction in Table VI. In this setting, the DA is removed. When considering only spatial interaction, structural overall and subtle detail changes cannot be simultaneously handled, leading to decreased performance. With the introduction of the spatial interaction, the change detection performance is also improved.

We further present the performance comparison with and without adaptive adjustment of each frequency component in Table VII. The WSFI module, with its enhanced frequency component selection mechanism, demonstrated superior performance across three datasets. This ablation experiment revealed that different scenarios have varying preferences for different frequency information. Dynamically adjusting the weights of these frequency components can significantly enhance the network's performance in change detection.

TABLE VI
IMPACT OF THE SPATIAL AND FREQUENCY INTERACTION

Dataset	Method	P	R	F1	IoU	OA
LEVIR-CD	w/o frequency	92.24	89.34	90.77	83.10	90.07
	w/o spatial	92.23	**89.59**	90.90	83.30	99.06
	Ours	**92.83**	89.25	**91.01**	**83.50**	**99.10**
WHU-CD	w/o frequency	95.07	91.26	93.13	87.14	99.41
	w/o spatial	94.58	**92.66**	93.66	87.99	**99.50**
	Ours	**96.58**	91.86	**94.16**	**88.97**	**99.50**
CLCD	w/o frequency	77.63	76.39	77.01	62.61	96.61
	w/o spatial	78.48	**77.50**	77.99	63.92	96.74
	Ours	**84.34**	74.29	**79.00**	**65.28**	**97.06**

3) Effectiveness of Semi-Supervised Domain Adaptation: In this section, we examine the effectiveness of the semi-supervised DA. Four cases are considered: without DA, without interclass domain discrepancy maximization in changed regions, without intraclass domain discrepancy minimization in unchanged regions, and our proposed method. Table VIII shows that incorporating the domain discrepancy learning for either changed or unchanged areas enhances performance. Benefiting from the contrastive DA in both regions, our method achieves the best performance. Fig. 8 presents the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:24:32 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
PERFORMANCE COMPARISON WITH/WITHOUT FREQUENCY COMPONENT ADJUSTMENT

Dataset	Method	P	R	F1	IoU	OA
LEVIR-CD	Without	91.95	**89.97**	90.95	83.40	99.09
	With	**92.83**	89.25	**91.01**	**83.50**	**99.10**
WHU-CD	Without	95.99	91.54	93.71	88.17	99.46
	With	**96.58**	**91.86**	**94.16**	**88.97**	**99.50**
CLCD	Without	83.70	72.89	77.92	63.83	96.93
	With	**84.34**	**74.29**	**79.00**	**65.28**	**97.06**

TABLE VIII
IMPACT OF DOMAIN ADAPTATION

Dataset	Method	P	R	F1	IoU	OA
LEVIR-CD	baseline	92.24	89.34	90.77	83.10	99.07
	w/o Change	92.01	90.13	91.06	83.58	99.10
	w/o Unchange	91.55	**90.45**	91.00	83.49	99.09
	Ours	**92.47**	89.77	**91.10**	**83.66**	**99.11**
WHU-CD	baseline	95.07	91.26	93.13	87.14	99.41
	w/o Change	95.82	91.51	93.62	88.00	99.45
	w/o Unchange	94.86	**91.94**	93.38	87.58	99.43
	Ours	**96.28**	91.79	**93.98**	**88.64**	**99.48**
CLCD	baseline	77.63	76.39	77.01	62.61	96.61
	w/o Change	79.60	**76.72**	78.13	64.11	96.80
	w/o Unchange	**84.63**	72.06	77.84	63.72	96.95
	Ours	83.33	74.32	**78.57**	**64.70**	**96.98**

TABLE IX
PERFORMANCE COMPARISON WITH/WITHOUT K-MEANS

Dataset	K-means	P	R	F1	IoU	OA
LEVIR-CD	Without	92.47	**90.34**	**91.39**	**84.15**	**99.13**
	With	**92.57**	90.08	91.31	84.01	**99.13**
WHU-CD	Without	**95.73**	93.24	94.48	89.55	**99.52**
	With	95.36	**93.70**	**94.52**	**89.61**	**99.52**
CLCD	Without	82.82	76.56	79.56	66.06	97.07
	With	**84.58**	**75.47**	**79.77**	**66.34**	**97.15**

feature distribution on the CLCD dataset. The red and blue dots represent the feature distributions of changed areas, while purple and green dots indicate the feature distributions of unchanged areas in two temporal feature maps. It is observed that the semi-supervised DA scheme brings closer the distributions of unchanged regions (e.g., green and purple points) while pushing away the distributions of changed regions (e.g., blue and red points). This phenomenon implies the effectiveness of incorporating semi-supervised DA.

Moreover, we performed ablation study with and without K-means for region division to show its effectiveness. As can be observed in Table IX, the performance improves in most cases with the K-means, especially in the complex WHU-CD and CLCD datasets.

V. CONCLUSION

This article presents a novel approach for remote sensing image change detection using a wavelet-based Siamese network with semi-supervised DA. The integration of wavelet-based spatial–frequency interaction facilitates accurate feature difference extraction, enabling simultaneous detection of overall changes and improved change boundary completeness. Leveraging semi-supervised DA can enhance the intraclass compactness within the unchanged regions and interclass separability between changed regions, effectively suppressing pseudochanges and enhancing the identification of targeted changes. The hybrid advantages of our method contribute to its superior performance compared with many competing methods. In future work, we plan to integrate foundational model knowledge to further enhance change detection capabilities.

REFERENCES

[1] A. Toker et al., "DynamicEarthNet: Daily multi-spectral satellite dataset for semantic change segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 21158–21167.

[2] Y. Li, J. Ren, Y. Yan, P. Ma, M. Assaad, and Z. Gao, "ABBD: Accumulated band-wise binary distancing for unsupervised parameter-free hyperspectral change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 9880–9893, 2024.

[3] K. Yang et al., "Asymmetric Siamese networks for semantic change detection in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5609818.

[4] L. Bergamasco, S. Saha, F. Bovolo, and L. Bruzzone, "Unsupervised change detection using convolutional-autoencoder multiresolution features," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408119.

[5] J. Shi, T. Wu, A. K. Qin, Y. Lei, and G. Jeon, "Semisupervised adaptive ladder network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5408220.

[6] C. Wu, B. Du, and L. Zhang, "Fully convolutional change detection framework with generative adversarial network for unsupervised, weakly supervised and regional supervised change detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 8, pp. 9774–9788, Aug. 2023.

[7] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[8] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.

[9] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Jul. 2022, Art. no. 5607514.

[10] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 207–210.

[11] Y. Zhong, B. Li, L. Tang, S. Kuang, S. Wu, and S. Ding, "Detecting camouflaged object in frequency domain," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 4504–4513.

[12] J. Liu, W. Xuan, Y. Gan, Y. Zhan, J. Liu, and B. Du, "An end-to-end supervised domain adaptation framework for cross-domain change detection," *Pattern Recognit.*, vol. 132, Dec. 2022, Art. no. 108960.

[13] C. Zhao et al., "High-resolution remote sensing bitemporal image change detection based on feature interaction and multitask learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5511514.

[14] Q. Liu, K. Ren, X. Meng, and F. Shao, "Domain adaptive cross reconstruction for change detection of heterogeneous remote sensing images via a feedback guidance mechanism," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4507216.

[15] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2018.

[16] M. Liu, Z. Chai, H. Deng, and R. Liu, "A CNN-transformer network with multiscale context aggregation for fine-grained cropland change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4297–4306, 2022.

[17] T. Li, F. Xiong, W. Zheng, Z. Li, J. Zhou, and Y. Qian, "Wavelet Siamese network for change detection in remote sensing images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2023, pp. 5455–5458.

[18] D. Marinelli, N. C. Coops, D. K. Bolton, and L. Bruzzone, "Forest change detection in LiDAR data based on polar change vector analysis," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:24:32 UTC from IEEE Xplore. Restrictions apply.

[19] M. Zanetti, F. Bovolo, and L. Bruzzone, "Rayleigh–Rice mixture parameter estimation via EM algorithm for change detection in multispectral images," *IEEE Trans. Image Process.*, vol. 24, no. 12, pp. 5004–5016, Dec. 2015.

[20] L. Bruzzone and D. F. Prieto, "An adaptive semiparametric and context-based approach to unsupervised change detection in multitemporal remote-sensing images," *IEEE Trans. Image Process.*, vol. 11, no. 4, pp. 452–466, Apr. 2002.

[21] J. S. Deng, K. Wang, Y. H. Deng, and G. J. Qi, "PCA-based land-use change detection and analysis using multitemporal and multisensor satellite data," *Int. J. Remote Sens.*, vol. 29, no. 16, pp. 4823–4838, Aug. 2008.

[22] Y. Sun, L. Lei, D. Guan, G. Kuang, and L. Liu, "Graph signal processing for heterogeneous change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4415823.

[23] B. Du, Y. Wang, C. Wu, and L. Zhang, "Unsupervised scene change detection via latent Dirichlet allocation and multivariate alteration detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 12, pp. 4676–4689, Dec. 2018.

[24] Y. Li, M. Gong, L. Jiao, L. Li, and R. Stolkin, "Change-detection map learning using matching pursuit," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 8, pp. 4712–4723, Aug. 2015.

[25] H. Nemmour and Y. Chibani, "Multiple support vector machines for land cover change detection: An application for mapping urban extensions," *ISPRS J. Photogramm. Remote Sens.*, vol. 61, no. 2, pp. 125–133, Nov. 2006.

[26] Y. Feng, J. Jiang, H. Xu, and J. Zheng, "Change detection on remote sensing images using dual-branch multilevel intertemporal network," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4401015.

[27] J. Wang, Y. Zhong, and L. Zhang, "Change detection based on supervised contrastive learning for high-resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5601816.

[28] H. Chen, C. Wu, B. Du, L. Zhang, and L. Wang, "Change detection in multisource VHR images via deep Siamese convolutional multiple-layers recurrent neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2848–2864, Apr. 2020.

[29] Y. Feng, H. Xu, J. Jiang, H. Liu, and J. Zheng, "ICIF-Net: Intra-scale cross-interaction and inter-scale feature fusion network for bitemporal remote sensing images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4410213.

[30] B. Jiang et al., "VcT: Visual change transformer for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 2005214.

[31] B. Bai, W. Fu, T. Lu, and S. Li, "Edge-guided recurrent convolutional neural network for multitemporal remote sensing image building change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5610613.

[32] F. Luo, T. Zhou, J. Liu, T. Guo, X. Gong, and J. Ren, "Multiscale diff-changed feature fusion network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5502713.

[33] Y. Wu et al., "CSTSUNet: A cross Swin transformer-based Siamese U-shape network for change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5623715.

[34] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[35] H. F. Tolie, J. Ren, and E. Elyan, "DICAM: Deep inception and channel-wise attention modules for underwater image enhancement," *Neurocomputing*, vol. 584, Jun. 2024, Art. no. 127585.

[36] Y. Li et al., "CBANet: An end-to-end cross-band 2-D attention network for hyperspectral change detection in remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5513011.

[37] J. Chen et al., "DASNet: Dual attentive fully convolutional Siamese networks for change detection in high-resolution satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1194–1206, 2021.

[38] X. Peng, R. Zhong, Z. Li, and Q. Li, "Optical remote sensing image change detection based on attention mechanism and image difference," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7296–7307, Sep. 2021.

[39] Y. Liu, C. Pang, Z. Zhan, X. Zhang, and X. Yang, "Building change detection for remote sensing images using a dual-task constrained deep Siamese convolutional network model," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 811–815, May 2021.

[40] X. Zhang, S. Cheng, L. Wang, and H. Li, "Asymmetric cross-attention hierarchical network based on CNN and transformer for bitemporal remote sensing images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 2000415.

[41] X. Tang, T. Zhang, J. Ma, X. Zhang, F. Liu, and L. Jiao, "WNet: W-shaped hierarchical network for remote-sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5615814.

[42] K. Li, X. Cao, and D. Meng, "A new learning paradigm for foundation model-based remote-sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5610112.

[43] L. Ding, K. Zhu, D. Peng, H. Tang, K. Yang, and L. Bruzzone, "Adapting segment anything model for change detection in VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5611711.

[44] L. Mou, L. Bruzzone, and X. X. Zhu, "Learning spectral–spatial–temporal features via a recurrent convolutional neural network for change detection in multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 924–935, Feb. 2019.

[45] G. Cheng, G. Wang, and J. Han, "ISNet: Towards improving separability for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5623811.

[46] Q. Li, R. Zhong, X. Du, and Y. Du, "TransUNetCD: A hybrid transformer network for change detection in optical remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5622519.

[47] L. Zhang, X. Hu, M. Zhang, Z. Shu, and H. Zhou, "Object-level change detection with a dual correlation attention-guided detector," *ISPRS J. Photogramm. Remote Sens.*, vol. 177, pp. 147–160, Jul. 2021.

[48] C. Zhang et al., "A deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 183–200, Aug. 2020.

[49] Y. Tang et al., "An object fine-grained change detection method based on frequency decoupling interaction for high-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5600213.

[50] H. Zheng et al., "HFA-Net: High frequency attention Siamese network for building change detection in VHR remote sensing images," *Pattern Recognit.*, vol. 129, Sep. 2022, Art. no. 108717.

[51] S. Yuan, R. Zhong, Q. Li, and Y. Dong, "MFGFNet: A multi-scale remote sensing change detection network using the global filter in the frequency domain," *Remote Sens.*, vol. 15, no. 6, p. 1682, Mar. 2023.

[52] D. Quan et al., "Efficient and robust: A cross-modal registration deep wavelet learning method for remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 4739–4754, 2023.

[53] S. Xiang, Q. Liang, and L. Fang, "Discrete wavelet transform-based Gaussian mixture model for remote sensing image compression," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 3000112.

[54] H. Liu et al., "Spatial-phase shallow learning: Rethinking face forgery detection in frequency domain," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 772–781.

[55] M. Zhou et al., "Spatial-frequency domain information integration for pan-sharpening," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2022, pp. 274–291.

[56] Y. Zhou, Y. Feng, S. Huo, and X. Li, "Joint frequency-spatial domain network for remote sensing optical image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5627114.

[57] D. Xue et al., "Triple change detection network via joint multifrequency and full-scale Swin-transformer for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4408415.

[58] Y. Zhu, L. Fan, Q. Li, and J. Chang, "Multi-scale discrete cosine transform network for building change detection in very-high-resolution remote sensing images," *Remote Sens.*, vol. 15, no. 21, p. 5243, Nov. 2023.

[59] N. Dräger, Y. Xu, and P. Ghamisi, "Backdoor attacks for remote sensing data with wavelet transform," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5613715.

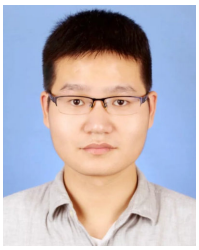
[60] B. Liu, L. Zhao, W. Liu, and Y. Li, "MWLN: Multilevel wavelet learning network for continuous-scale remote-sensing image super-resolution," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[61] G. Wilson and D. J. Cook, "A survey of unsupervised deep domain adaptation," *ACM Trans. Intell. Syst. Technol.*, vol. 11, no. 5, pp. 1–46, Oct. 2020.

[62] M. Wang and W. Deng, "Deep visual domain adaptation: A survey," *Neurocomputing*, vol. 312, pp. 135–153, Oct. 2018.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:24:32 UTC from IEEE Xplore. Restrictions apply.

[63] M. Long, H. Zhu, J. Wang, and M. I. Jordan, "Unsupervised domain adaptation with residual transfer networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2016, pp. 136–144.
[64] Y. Ganin and V. Lempitsky, "Unsupervised domain adaptation by backpropagation," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2015, pp. 1180–1189.
[65] M. Ghifary, W. B. Kleijn, M. Zhang, D. Balduzzi, and W. Li, "Deep reconstruction-classification networks for unsupervised domain adaptation," in *Proc. Eur. Conf. Comput. Vis.*, 2016, pp. 597–613.
[66] C. Zhang et al., "A domain adaptation neural network for change detection with heterogeneous optical and SAR remote sensing images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 109, May 2022, Art. no. 102769.
[67] T. Y. Lin, P. Dollàr, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 2117–2125.
[68] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, vol. 6, 2017.



Fengchao Xiong (Member, IEEE) received the B.E. degree in software engineering from Shandong University, Jinan, China, in 2014, and the Ph.D. degree from the College of Computer Science, Zhejiang University, Hangzhou, China, in 2019.

He visited Wuhan University, Wuhan, China, from 2011 to 2012; Griffith University, Nathan, QLD, Australia, from 2017 to 2018; and the University of Macau, Macau, China, from 2021 to 2023. He is currently an Associate Professor with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing, China. His research interests include hyperspectral image processing, machine learning, and pattern recognition.

Dr. Xiong serves as a Topical Associate Editor for IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.



Tianhan Li received the B.E. and M.S. degrees in computer science and technology from Nanjing University of Science and Technology, Nanjing, China, in 2021 and 2024, respectively.

His research interests include machine learning and remote sensing image analysis.



Yi Yang received the B.E. and M.E. degrees from Huazhong University of Science and Technology, Hubei, China, in 2009 and 2012, respectively.

He is currently an Associate Researcher with the Natural Resources Investigation and Monitoring Research Center, Chinese Academy of Surveying and Mapping, Beijing, China. His research interests include remote sensing image processing and deep learning.



Jun Zhou (Senior Member, IEEE) received the B.S. degree in computer science and the B.E. degree in international business from Nanjing University of Science and Technology, Nanjing, China, in 1996 and 1998, respectively, the M.S. degree in computer science from Concordia University, Montreal, QC, Canada, in 2002, and the Ph.D. degree from the University of Alberta, Edmonton, AB, Canada, in 2006.

In June 2012, he joined the School of Information and Communication Technology, Griffith University, Nathan, QLD, Australia, where he is currently a Professor. Prior to this appointment, he was a Research Fellow at the Research School of Computer Science, Australian National University, Canberra, ACT, Australia, and a Researcher with the Canberra Research Laboratory, NICTA, Canberra. His research interests include pattern recognitions, computer vision, and spectral imaging with their applications in remote sensing and environmental informatics.

Dr. Zhou is an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and *Pattern Recognition Journal*.



Jianfeng Lu (Member, IEEE) received the M.S. and Ph.D. degrees in pattern recognition and intelligent system from the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing, China, in 1994 and 2000, respectively.

He is currently a Professor with the School of Computer Science and Engineering, Nanjing University of Science and Technology. His research interests include image analysis, machine learning, data mining, and intelligent robot.



Yuntao Qian (Senior Member, IEEE) received the B.E. and M.E. degrees in automatic control from Xi'an Jiaotong University, Xi'an, China, in 1989 and 1992, respectively, and the Ph.D. degree in signal processing from Xidian University, Xi'an, in 1996.

From 1996 to 1998, he was a Post-Doctoral Fellow with Northwestern Polytechnical University, Xi'an. Since 1998, he has been with the College of Computer Science, Zhejiang University, Hangzhou, China, where he became a Professor in 2002. He was a Visiting Professor with Concordia University, Montreal, QC, Canada, from 1999 to 2001; Hong Kong Baptist University, Hong Kong, in 2006; Carnegie Mellon University, Pittsburgh, PA, USA, in 2010; the Canberra Research Laboratory, NICTA, Canberra, ACT, Australia, in 2013; Macau University, Macau, China, from 2015 to 2016; and Griffith University, Nathan, QLD, Australia, in 2018. His research interests include machine learning, signal and image processing, pattern recognition, and hyperspectral imaging.

Dr. Qian is currently an Associate Editor of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:24:32 UTC from IEEE Xplore. Restrictions apply.